7th April 2026

External Project: Mid - Term Progress Presentation

Self-Supervised Temporal Ultrasound Pose Estimation under Limited Supervision for Trackerless Freehand 3D Reconstruction

Presented By

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Presentation Outline

Introduction

Reconstructing 2D Ultrasound (US) images into a 3D volume enables 3D representations of anatomy to be generated which are beneficial to a wide range of downstream tasks such as quantitative biometric measurement, multimodal registration, and 3D visualisation.

But in tracker-free freehand US, the challenge is to estimate relative transformations between frames directly from the ultrasound data itself (e.g. speckle decorrelation, optical flow, deep learning)

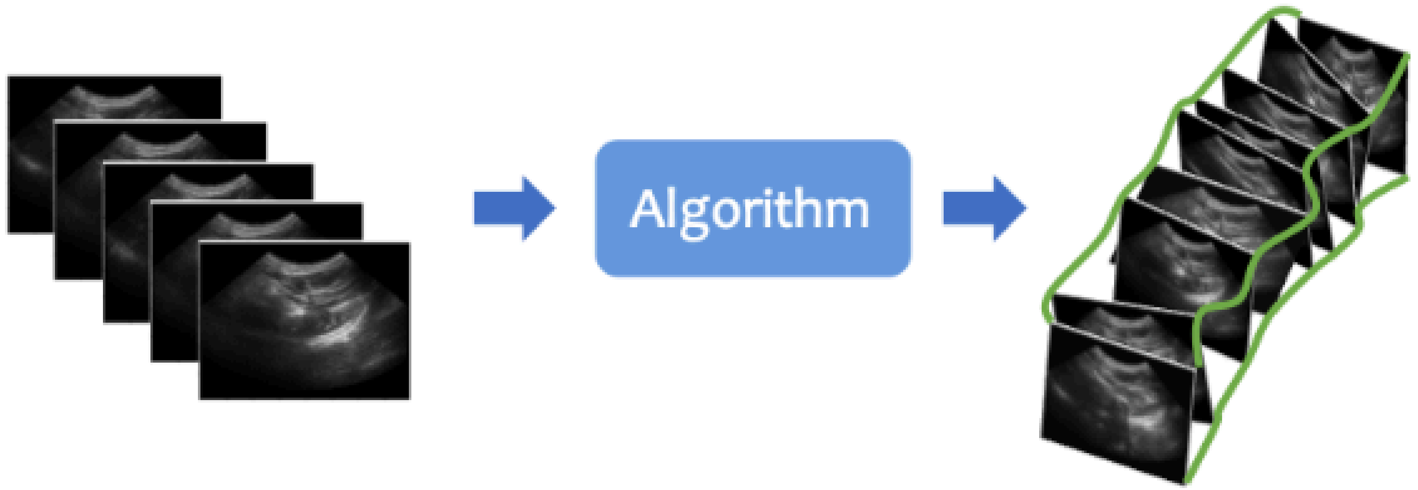

Fig 1: An Illustration of a freehand US reconstruction (**Source**)

Source: https://github-pages.ucl.ac.uk/tus-rec-challenge/

Literature Survey/Background Theory

Article Title	Authors & Year	Core Idea (Short)	Datasets	Key Results	Tools / Sensors Used
3D Freehand Ultrasound without Tracking	Prevost et al., 2018	Deep learning for trackerless reconstruction	Ultrasound sequences	Landmark work in sensorless US	Ultrasound probe
Geometric Consistency for Visual Odometry	Iyer et al., 2018	Self-supervised visual odometry	KITTI dataset	Improved pose estimation	Cameras
A Simple Framework for Contrastive Learning (SimCLR)	Chen et al., 2020	Contrastive self-supervised learning framework	ImageNet	State-of-the-art representation learning without labels	RGB images
Quaternions and Rotations	Jia, n.d.	Mathematical foundation for rotations	N/A	Widely used in 3D transformations	Mathematical framework
Survey on 3D Ultrasound Reconstruction	Mohamed & Vei Siang, 2019	Overview of reconstruction methods	Various	Categorization of techniques	Ultrasound
Temporal Cycle-Consistency Learning	Dwibedi et al., 2019	Self-supervised temporal consistency	Video datasets	Improved representation learning in videos	Video sequences
Sensorless Freehand 3D Ultrasound Reconstruction	Guo et al., 2020	Deep contextual learning for reconstruction	Ultrasound data	Accurate 3D reconstruction without tracking	Ultrasound probe
Self-supervised Contrastive Learning for Time-Series	Eldele et al., 2023	Contrastive learning for time-series	Time-series datasets	Strong semi-supervised performance	Time-series sensors
Sensorless End-to-End Freehand Ultrasound	Dou et al., 2023	Physics-inspired network for reconstruction	Ultrasound sequences	Improved reconstruction without trackers	Ultrasound probe
Trackerless Freehand Ultrasound (Sequence Modeling)	Li et al., 2023	Sequence modeling for reconstruction	Ultrasound sequences	Better temporal consistency	Ultrasound probe
Temporal Supervised Contrastive Learning	Noroozizadeh et al., n.d.	Contrastive learning for tabular time-series	Tabular datasets	Improved classification	Time-series data

Baseline Papers - Used and referenced heavily

Problem Definition

"Build a model that can estimate relative ultrasound probe poses (translation + rotation) from a short temporal window of B-mode frames, so those predicted poses can be integrated into a trajectory and used for 3D reconstruction."

Motivation for Trackerless Approach

Tracking Modality	Pros	Cons	Key References
Electromagnetic (EM) Trackers	• Provide real-time 6-DoF tracking. • Do not require line-of-sight. • Widely used in freehand 3D ultrasound and image-guided interventions.	• Highly susceptible to magnetic field distortions from metallic objects and clinical equipment. • Reduced accuracy in complex hospital environments. • Require careful calibration and controlled setups.	Hummel et al., 2005; Franz et al., 2014; Mercier et al., 2005
Optical Trackers	• High spatial accuracy and precision. • Considered a gold standard for motion tracking. • Immune to electromagnetic interference.	• Strict line-of-sight requirements. • Easily occluded by clinicians' hands or surgical instruments. • Bulky hardware disrupts clinical workflows. • Requires frequent hand–eye calibration.	Wiles et al., 2004; Mercier et al., 2005; Mountney & Yang, 2010
Inertial Measurement Units (IMUs)	• Compact, lightweight, and cost-effective. • Capture orientation reliably. • Easily integrable into portable devices.	• Position estimation requires double integration of acceleration. • Accumulation of noise leads to exponential drift. • Unsuitable for accurate long-duration trajectory reconstruction without external correction.	Woodman, 2007; Titterton & Weston, 2004; Foxlin, 2005

Motivation - Image Driven Trajectory Estimation

Tracking Modality	Pros	Cons	Key References
Deep Learning-Based Image Tracking	• Immune to magnetic distortions and visual occlusions. • Eliminates the need for external tracking hardware. • Enables software-defined 3D reconstruction from 2D B-mode ultrasound. • Facilitates point-of-care diagnostics. • Preserves probe ergonomics and workflow efficiency.	• Requires large, high-quality datasets for training. • Computationally intensive. • May face generalization challenges across devices and clinical settings. • Limited interpretability compared to physics-based methods.	Prevost et al., 2017; Hou et al., 2021; Salehi et al., 2018; Haskins et al., 2020

Objective

- **Develop a Self-Supervised Pipeline:** Estimate relative ultrasound probe poses directly from image sequences to minimize reliance on external tracking data and dense annotations.
- **Enhance Trajectory Stability:** Enforce geometric and temporal consistency constraints like cycle consistency and composition consistency to physically ground the predicted motion and significantly reduce drift*.
- **Enable Trackerless 3D Reconstruction:** Integrate predicted frame-to-frame motion into a stable trajectory to generate accurate 3D ultrasound volumes without external hardware.

*

translation part of matrix $\boldsymbol{T}_k$, we can furthermore compute the *final drift* of a sweep as the distance between the positions of the last frame (N) center of the estimated trajectory and the ground truth trajectory:

$$\text{drift} = \|\vec{t}_N - \vec{t}_N^{GT}\|. \quad (6)$$

In other words, this number is a target registration error on the center of the final frame and captures the accumulated error over the whole sweep. The *maximum center error* is the maximum dis-

Fig 2: Drift Definition (Snippet From Prevost's Paper) (**Source**)

Source: Prevost, R., Salehi, M., Jagoda, S., Kumar, N., Sprung, J., Ladikos, A., Bauer, R., Zettinig, O., & Wein, W. (2018). 3D freehand ultrasound without external tracking using deep learning. Medical Image Analysis, 48, 187–202.

Dataset Overview

Dataset is a sequence-based, where each sequence contains:

- 2D ultrasound frames: *.jpg
- **Tracker pose file:** Inputforrecon.mha with per-frame 4x4 probe transforms

Raw Input Type

For each usable frame:

- Image is loaded as grayscale float in [0,1]
- Cropped ROI is used
- Final frame tensor shape is (1, H, W)

Tracker ground-truth source is MHA:

Each frame ID maps to a **4x4 absolute SE(3) transform**

Dataset Overview

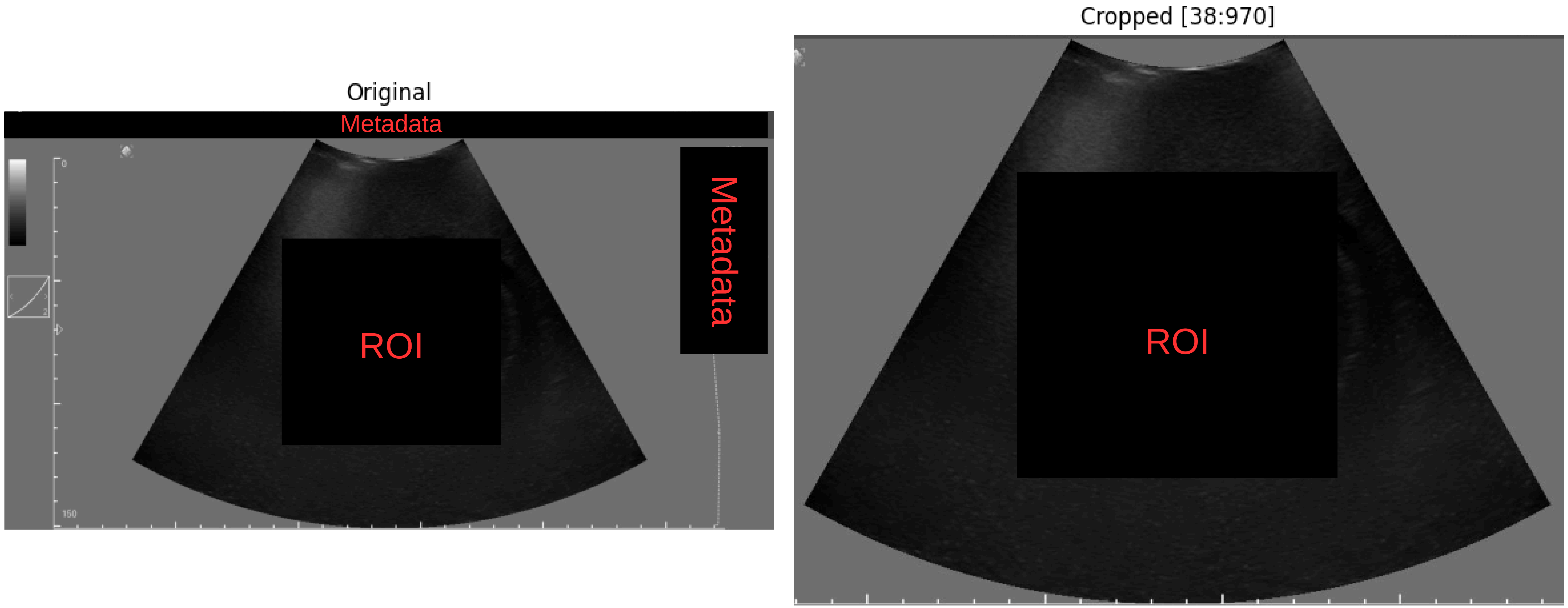

Fig 3: Cropped ROI (Generated from phantom dataset) **- UTSAAH Lab**

Note: Part of the image has been redacted due to an **NDA** signed between the employee (**Abeer Sethia**) and employer (**UTSAAH Lab**).

Dataset Overview

Model Inputs and Ground Truth (two datasets)

1) SSL dataset (PhantomTemporalPairDataset)

- **Input to encoder**: pair (x1, x2) of temporally close frames
- **Expected output**: no explicit pose target; used for contrastive loss only

2) Pose dataset (PhantomPoseWindowDataset)

- **Input**: a temporal window of frames (default odd window like 5), shape (W, 1, H, W)
- **Ground truth**: one relative pose per window
  - t_rel: translation (3,)
  - q_rel: quaternion (4,)

GT is computed from tracker transforms of idx-1 and idx+1 (temporarily)

Methodology

A two-stage temporal ultrasound framework is used: first, a self-supervised contrastive encoder **learns robust frame features from unlabeled sequences**; second, a temporal pose regressor **predicts frame-to-frame relative poses from short feature windows**, trained with supervised pose targets plus geometric consistency constraints like **cycle consistency and CTC (Composite Transformation Constraints).** The predicted relative poses are then integrated into a trajectory for downstream 3D reconstruction/evaluation.

Note: A detailed methodology cannot be provided due to an **NDA** signed between the intern (**Abeer Sethia**) and employer (**UTSAAH Lab**).

Methodology

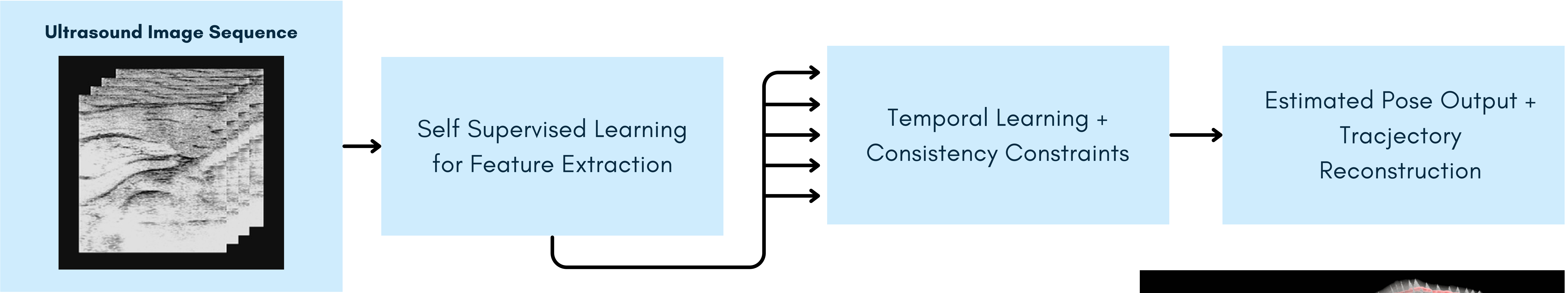

Fig 4: Ultrasound Image Sequence (Source)

Temporal Window Creation
Example: Windows of length 5
(Features Extracted From Image 1...5)

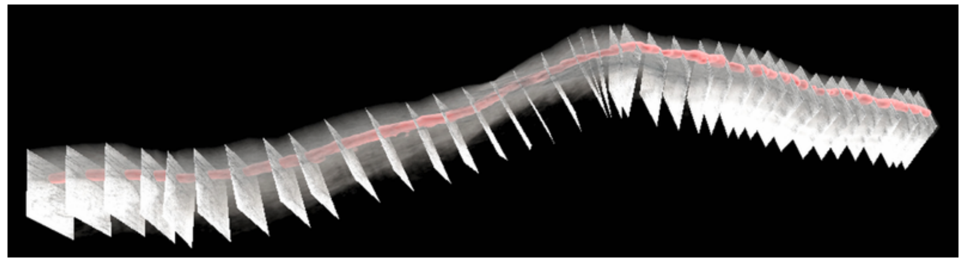

Fig 5: Reconstruction of the ultrasound sequence with the red line indicating trajectory (**Source**)

Note: A detailed methodology cannot be provided due to an **NDA** signed between the intern (**Abeer Sethia**) and employer (**UTSAAH Lab**).

Constraints and Limitations

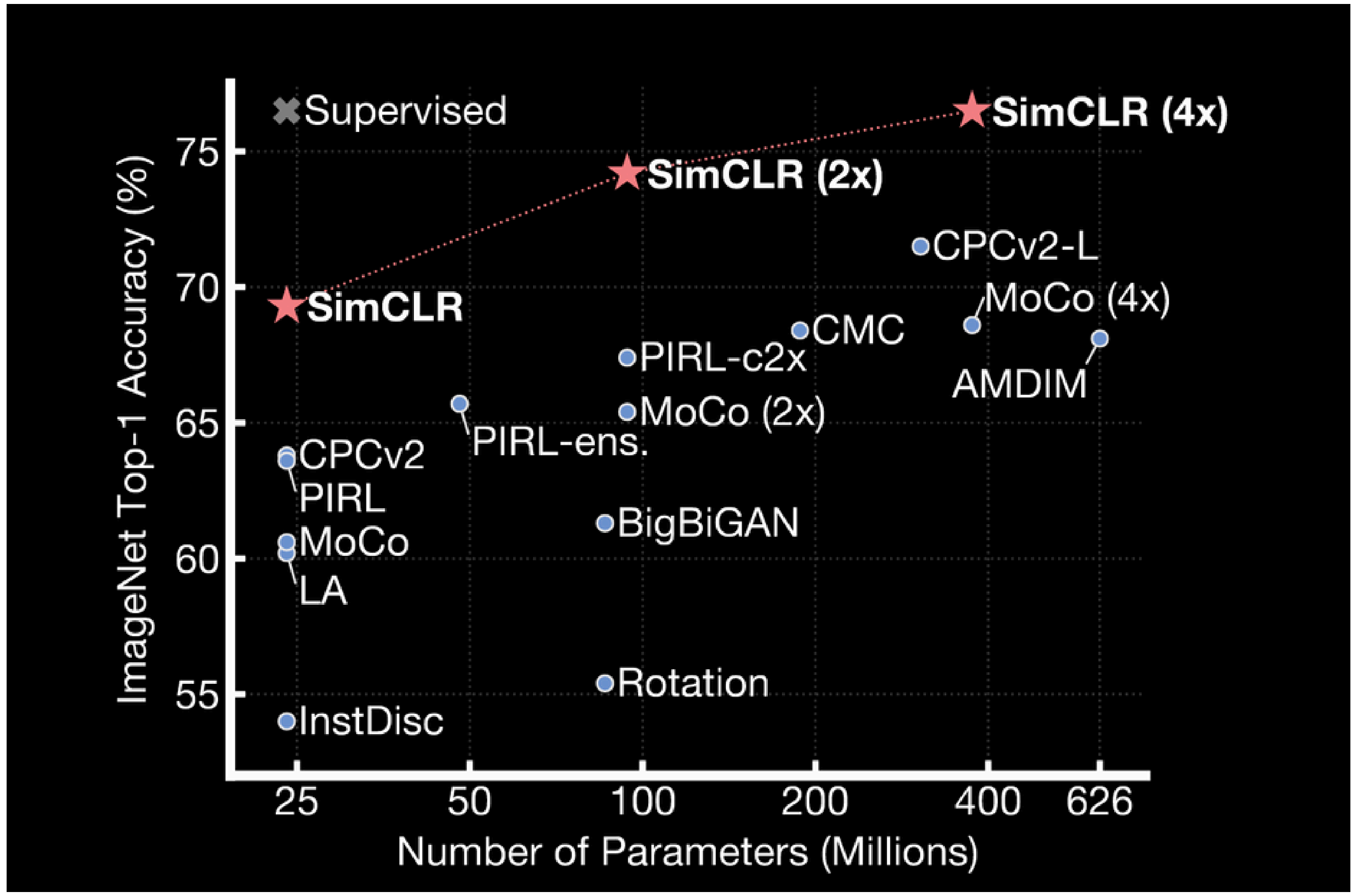

Fig 6: Supervised v.s Other Approaches for Contrastive Learning. SimCLR: A simple framework for contrastive learning of visual representations (**Source**)

While existing literature indicates that self-supervised learning (SSL) approaches generally underperform compared to fully supervised baselines in absolute accuracy, the adoption of SSL is driven by a critical practical necessity: the ability to leverage abundant, unannotated datasets in domains where acquiring ground-truth labels is prohibitively expensive or unfeasible.

Note: A detailed explanation on the limitations cannot be provided due to an **NDA** signed between the intern (**Abeer Sethia**) and employer (**UTSAAH Lab**).

Work Done So Far

The project began with Problem Understanding to define core objectives, followed by establishing the foundational Theory to guide the approach. We then completed the Geometry Implementation to handle necessary spatial modeling and built a custom Data Loader for efficient data ingestion. Currently, we are actively developing the Pipeline (in progress) to integrate these individual components into a single, end-to-end workflow.

Remaining Work

The remaining work focuses on system validation and project finalization. First, Pipeline Testing will be conducted to ensure the fully integrated, end-to-end workflow operates reliably and without errors. Next, Ablation Studies will be performed to systematically evaluate the impact and necessity of individual system components on overall performance. Finally, comprehensive Documentation will be compiled to clearly detail the methodology, codebase, and results for future reference and reproducibility.

Scope

- Develop and validate a learning-based temporal ultrasound motion estimation pipeline.
- Use feature learning to reduce dependence on dense manual annotations.
- Train a relative pose model on short frame windows using available tracking-derived targets.
- Enforce geometric/temporal consistency during training for physically plausible motion.
- Reconstruct and evaluate trajectory-driven 3D volume formation from predicted motion.

Conclusion

In conclusion, the project has successfully established its theoretical foundation and core functional components, and is now actively transitioning into the mid stages of system integration. Once the pipeline is fully assembled, the focus will shift to rigorous validation through comprehensive testing and ablation studies. These final evaluation steps, culminating in detailed documentation, will ensure the delivery of a robust, well-verified, and fully reproducible system.

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DEPARTMENT OF
DESIGN AND MANUFACTURING
INDIAN INSTITUTE OF SCIENCE

UTSAAH
LAB

16th December 2025

NON-DISCLOSURE AGREEMENT

THIS AGREEMENT FOR NONDISCLOSURE OF CONFIDENTIAL INFORMATION (the "Agreement") is made and entered into as of 10th December 2025 between UTSAAH Lab, Department of Design and Manufacturing, Indian Institute of Science, Bangalore, **(hereinafter referred to as the "Hiring Party")**

AND

Abeer Sethia a permanent resident of #1D, Tejpunj, 20/1, Venkateshwaranagar, Jakkur, Bangalore - 560064 **(hereinafter referred to as "Associate"),** Together the Hiring Party and the Associate shall constitute and be hereinafter referred to as the **"Party" or "Parties"**.

1. Purpose:

Hiring Party has hired the Associate as **Research Intern - AI** starting from **10th December 2025 till 10th June 2026 (duration of 06 months).**

and the Hiring Party will therefore disclose to Associate certain confidential technical and business information, which the Hiring Party desires the Associate to treat as confidential and proprietary**("Purpose")**.

2. Confidential Information:

"Confidential Information" means any information disclosed to Associate by the Hiring Party, either directly or indirectly in writing, by delivery of items, by initiation of access such as maybe in a data base, electronically, by oral or by visual presentation or inspection of tangible objects, including without limitation documents, business plans, product mock ups, content, use cases and stories, designs, methodologies, algorithms, processes, source code, documentation, financial analysis, marketing plans, customer names, customer list, customer data, which is marked "Confidential" or "Proprietary", or which by its nature or the circumstances or manner of its disclosure would logically be considered proprietary or confidential.

Confidential Information may also include information disclosed to the Hiring Party by third parties. Confidential Information shall not, however, include any information which the Associate can establish:

(i) was publicly known and made generally available in the public domain prior to the time of disclosure to the Associate by the Hiring Party; or

(ii) becomes publicly known and made generally available after disclosure to the Associate by the Hiring Party through no action or inaction or fault of the Associate; or

(iii) is in the possession of Associate, without confidentiality restrictions, at the time of disclosure by Hiring Party as shown by Associate's files and records immediately prior to the time of disclosure.

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Bangalore - 560012
Email: marora@iisc.ac.in

3. Non-use, Non-disclosure, and Non-conflicting Obligations:

Associate agrees not to use any Confidential Information for any purpose whatsoever except to perform work for the Hiring Party as defined and/or required by the Hiring Party from time to time.

Hiring Party has hired the Associate as **Research Intern - AI** and the Hiring Party will therefore disclose to Associate certain confidential technical and business information, which the Hiring Party desires the Associate to treat as confidential and proprietary. Associate agrees not to disclose any Confidential Information to third parties or to colleagues/employees/family/friends/others.

Associate shall not, for Associate or third party, reverse engineer, repurpose, create derivatives, disassemble, or decompile any designs, content, processes, models, algorithms, prototypes, software, or other tangible objects which embody Hiring Party's Confidential Information, and which are provided to Associate hereunder. The Associate certifies that Associate has no outstanding agreement or obligation that is in conflict with any of the provisions of this Agreement and the Associate further agrees not to enter into any agreement which will conflict with the Agreement or prohibit the Associate from carrying out the obligations under this Agreement.

4. Maintenance of Confidentiality:

Associate agrees that he/she shall take all reasonable measures to protect the secrecy of and avoid disclosure and unauthorized use of the Confidential Information. Without limiting the foregoing, the Associate shall use at least the same care and use the same discretion to avoid disclosure, publication, or dissemination of the Hiring Party's Confidential Information, as the Associate uses with its own most highly secret and confidential information. The Associate shall not make any copies of Confidential Information unless the same are previously approved in writing by the Hiring Party. The Associate shall reproduce the Hiring Party's proprietary rights notices on any such approved copies, in the same manner in which such notices were set forth in or on the original. The Associate shall immediately notify the Hiring Party in the event of any unauthorized use or disclosure of the Confidential Information.

In order to comply with the provisions of this Agreement, and in order to preserve the confidential nature of the Confidential Information that the Hiring Party may disclose to the Associate pursuant to this Agreement, the Associate agrees it will employ at least the following procedures:
a) all electronic Confidential Information will be stored and accessed on a password protected computer whose password is personal;
b) all documents will only be on this password protected computer;
c) all non-electronic Confidential Information will be stored in locked filing cabinets for which, access is restricted to those authorized under this Agreement to receive such Confidential Information; and
d) all electronic Confidential Information disclosed by Hiring Party will be stored in a secure, protected file with access to such file via private password only and access to such electronic Confidential Information will be restricted to those authorized under this Agreement to receive such information.

5. No Obligation:

The Hiring Party reserves the right, in its sole discretion, to terminate this Agreement concerning the employment at any time. The Associate may terminate the agreement with a written notification to the Hiring Party by providing 02 months' notice. All terms related to Confidential

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Information will survive the lifetime of the agreement. All terms related to conflicts of interest will be applicable for 24 months from the date of termination of the agreement.

6. No Warranty:

All confidential information is provided "AS IS". The Hiring Party makes no warranties, express, implied, or otherwise, regarding its accuracy, completeness, or performance.

7. Destruction or Return of Materials:

All documents and other tangible objects in any form provided to Associate or prepared by Associate in connection with the use of the Confidential Information containing or representing Confidential Information and all copies, or extracts thereof, which are in the possession of Associate shall be and remain the property of the Hiring Party and shall be promptly returned to the Hiring Party or destroyed, at the Hiring Party's discretion, upon the Hiring Party's request. The Associate will furnish the Hiring Party with written confirmation of any such destruction.

8. No License:

Neither this Agreement nor any disclosure of Confidential Information that may be disclosed by Hiring Party under this Agreement grants any rights or license to Associate under any United States and/or international trademark, copyright, or patent, mask work right, and/or any application(s) thereof, now, or subsequently invented, owned and/or controlled by the Hiring Party.

9. Term:

This Agreement shall survive until such time as all Confidential Information disclosed hereunder becomes publicly known and made generally available through no action or inaction or fault of the Associate. Any terms of this Agreement which by their nature extend beyond its termination, remain in effect until fulfilled. This will constitute the lifetime of the agreement.

10. Dispute:

This Agreement shall be governed by, construed, and enforced in accordance with the laws of India.

11. Arbitration:

Any dispute, controversy, or claim arising out of or relating to this Agreement, or the breach there of, shall be settled by the jurisdictional courts at Bangalore, India.

12. Remedies:

The Associate agrees that any violation or threatened violation of this Agreement will cause irreparable injury to the Hiring Party, entitling Hiring Party to obtain injunctive relief in addition to all legal remedies.

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13. Associate Information:

The Hiring Party does not wish to receive any confidential information from the Associate, and the Hiring Party assumes no obligation, either express or implied, with respect to any information disclosed by Associate.

14. Miscellaneous:

This Agreement shall bind and inure to the benefit of the parties hereto. The Associate may not assign, or otherwise transfer, its rights or delegate its duties or obligations under this Agreement without prior written consent of the Hiring Party. Any attempt to do so is void. Both the Hiring Party and the Associate consent to the application of the laws of the State of Karnataka and India, to govern, interpret, and enforce all rights, duties, and obligations arising from, or relating in any manner to, the subject matter of this Agreement, without regard to conflict of law principles. This Agreement is the complete and exclusive Agreement regarding the Hiring Party's disclosures of Confidential Information and replaces any prior oral or written communications between the parties regarding these disclosures. Any failure to enforce any provision of this Agreement shall not constitute a waiver thereof or of any other provision hereof.

This Agreement may not be amended, nor any obligation waived, except by a writing signed by both parties hereto. By signing below for our respective companies by authorized signers, each of us agrees to the terms of this Agreement. Once signed, any reproduction of this Agreement made by reliable means is considered an original.

Signed By:

Name: Abeer Sethia

Title: Research Intern - AI

Date: 16.12.2025